EXP NO - 09 : CONTIGUOUS MEMORY ALLOCATION

FIRST FIT

```c
#include<stdio.h>
int main(){
        int m,n;
        printf("Enter number of memory blocks\n");
        scanf("%d",&m);
        int blockSize[m],remainingSize[m];
        printf("Enter size of each block:\n");
        for(int i=0;i<m;i++)
        {
                scanf("%d",&blockSize[i]);
                remainingSize[i]=blockSize[i];
        }
        printf("Enter number of processes:\n");
        scanf("%d",&n);
        int processSize[n];
        printf("Enter size of each process:\n");
        for(int i =0;i<n;i++)
                scanf("%d",&processSize[i]);

       for(int i=0;i<n;i++){
               int allocated =0;
               for(int j=0;j<m;j++){
                       if(remainingSize[j]>=processSize[i]){
                               int fragment = remainingSize[j]-processSize[i];
                               printf("Process %d of size %d is allocated to block %d of size %d with fragment %d\n",i+1,processSize[i],j+1,blockSize[j],fragm
ent);
                               remainingSize[j]-=processSize[i];
                               allocated=1;
                               break;
                }
               }
               if(!allocated)
                {
                        printf("process %d of size %d is not allocated \n",i+1,processSize[i]);
                }
        }
        return 0;
}

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"firstfit.c" 38L, 997B                                                                                                  23,1-8        All
```

```
[rene@localhost ~]$ vi firstfit.c
[rene@localhost ~]$ gcc firstfit.c
[rene@localhost ~]$ ./a.out
Enter number of memory blocks
5
Enter size of each block:
100
500
200
300
600
Enter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to block 2 of size 500 with fragment 288
Process 2 of size 417 is allocated to block 5 of size 600 with fragment 183
Process 3 of size 112 is allocated to block 2 of size 500 with fragment 176
process 4 of size 426 is not allocated
[rene@localhost ~]$
```